HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 117 : Engineering Ethics and Cyber Law

Batch: 27th | Semester: Fall 2024

Date: 04/01/2025 | **Midterm Examination** | Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions. $5 \times 6 = 30$

1. (a)	What is Moral Dilemmas?	CLO1	3
(b)	Briefly explain the types of complexities of Moral Dilemmas.	CLO1	3

2. John works for a cybersecurity firm that develops security software. A major client has requested a rush certification of their new banking application, offering to pay triple the usual fee. During testing, John finds several minor security vulnerabilities that technically pass the minimum requirements but could potentially be exploited. The client insists these can be patched in future updates after the launch.

(a)	Identify the ethical dilemma and explain how codes of professional ethics apply to this situation	CLO2	3
(b)	Using the concept of professional responsibility and integrity, what should John do? Justify your answer	CLO2	3

3. Think about a scenario, You are giving exam on this course and you didn't study the chapters properly. So naturally you are biting your pen instead of writing by it on the paper. Now suddenly you see your that same friend who told you last night that "Mama, kichuuiiii Pori nai. Kal Sure fail" is writing smoothly at the speed of light!! You also notice that the secret of his success is not his study rather his ability to do copy and bring 'Nokol' like a pro in the exam hall. — CLO2 — 6

Now explain this real-life part of yours from the 3 normative ethical approaches that we discussed on the class.

4. (a)	Outline the differences between morality and ethics	CLO2	3
(b)	Explain about customs and ethical relativism.	CLO2	3
5.	Explain the Lawrence Kohlberg's Theory on moral development.	CLO2	6
6. (a)	What is integrity in professional ethics? How does it relate to honesty and self-respect?	CLO2	3
(b)	Discuss the positive roles of codes of ethics in professional practice.	CLO1	3

what are its limitations?

7. Alex, a systems engineer, discovers that his company's new IoT device collects more personal user data than disclosed in their privacy policy. The data is being sold to third parties without user consent. When he raises this concern, his managers argue that all tech companies do this and it's essential for their business model.

(a)	Evaluate this situation using the three types of inquiries.	CLO2	3
(b)	Under what conditions whistleblowing would be ethically justified in this case? Support your answer with ethical principles.	CLO2	3

HAMDARD UNIVERSITY BANGLADESH

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Faculty of Science, Engineering and Technology

Course Code: MAT115 **Course Title: Mathematics-I (Differential and Integral Calculus)**
Mid-Term Examination Semester: Fall 2024
Date: 1/6/2025 Total Time: 1 Hour 30 Minutes **Total Marks: 30**

<u>Answer any three (03) out of the following questions</u>: $3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

			CLOs	Marks
Q1.	(a)	Define continuity of a function at a point $x = a$.	CLO1	[1]
	(b)	A function $f(x)$ is defined by $$f(x) = \begin{cases} 5x - 4\,;\ 0 < x \le 1 \\ 4x^2 - 3x\,;\ 1 < x < 2 \\ 3x + 4\,;\ x \ge 2 \end{cases}$$ Check the continuity of $f(x)$ at $x = 1$ and $x = 2$.	CLO3	[5]
	(c)	A function $f(x)$ is defined by $$f(x) = \begin{cases} x^2\,;\ x \le 0 \\ x\,;\ 0 < x < 1 \\ 2 - x\,;\ x \ge 1 \end{cases}$$ Check for the existence of limit at $x = 0$ and $x = 1$.	CLO3	[4]
Q2.	(a)	Define differentiability of a function at a point $x = a$.	CLO1	[1]
	(b)	A function $f(x)$ is defined as $$f(x) = \begin{cases} 1\,;\ x < 0 \\ 1 + \sin x\,;\ 0 \le x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2\,;\ x \ge \frac{\pi}{2} \end{cases}$$ Show that the function $f(x)$ is differentiable at $x = \frac{\pi}{2}$ but not at $x = 0$.	CLO3	[6]
	(c)	Determine $\frac{dy}{dx}$ for the followings (any one): (i) $y = (\tan x)^{\cot x} + (\cot x)^{\tan x}$ (ii) $x = a\cos^3\theta,\ \ y = a\sin^3\theta$	CLO3	[3]
Q3.	(a)	State and prove Leibnitz theorem.	CLO2	[5]
	(b)	If $y = e^{a\cos^{-1}x}$, then show that (i) $(1 - x^2)y_2 - xy_1 - a^2y = 0$ (ii) $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + a^2)y_n = 0$	CLO3	[5]
Q4.	(a)	Define an inflection point.	CLO1	[1]
	(b)	Justify whether the function $f(x) = x^3 - 3x^2 + 6x + 3$ has an inflection point or not. If exists, then determine it.	CLO2	[3]
	(c)	If $f(x) = \frac{x}{\ln x}$, then show that the minimum value of the function is e.	CLO3	[3]

	(d)	Determine the equation of tangent and normal for the curve $x^2 + y^2 - 4x + 3y - 2 = 0$ at the point $(-2,2)$.	CLO3	[3]
Q5.	(a)	Define partial differentiation of a function of two variables.	CLO1	[1]
	(b)	If $f(x,y) = \ln(x^2 + y^2)$, then show that $f_{xx} + f_{yy} = 0$.	CLO3	[3]
	(c)	State Euler's theorem for homogeneous functions of three variables.	CLO2	[2]
	(d)	Verify Euler's theorem for the function $f(x,y,z) = x^4 + y^4 + z^4 - x^2yz - xy^2z - xyz^2$	CLO3	[4]

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science and Engineering
Department of Computer Science and Engineering
Bachelor of Science in Computer Science and Engineering
ENG111-English Basics
Mid-Term Examination
Fall 2024

Batch 27th
January 08, 2025

1st Semester
Time: 1 Hour 30 Minutes

Total Marks: 30

Section A

Read the passage carefully and answer question 1 to 3 **CLO5**

Last weekend, I decided to visit my favorite park to enjoy the beautiful weather. The sun was shining brightly, and there were few clouds in the sky. As I walked along the main path, I saw several people sitting on benches, reading books, or chatting with friends. The park was very busy, as it was the weekend, but it still felt peaceful. A family was having a picnic near the lake, while children were running around playing games and their laughter was echoing through the air. In the distance, I could see a couple of people riding bicycles, while others were walking their dogs or jogging along the trail. There were groups of people playing frisbee, some of them laughing as they made excellent catches. As I passed by the playground, a group of kids were laughing and playing on the swings, pushing each other higher with every swing. I decided to stop and take a break under a large oak tree, where I could watch the activities around me. The breeze was gentle, and I could hear the sound of leaves rustling above. A friendly dog came up to me and began wagging its tail, looking for attention. It sat by my side, looking up at me with big, brown eyes. I petted the dog for a few minutes before continuing my walk. It was a peaceful afternoon, and I felt relaxed, enjoying the simple pleasures of nature. After a while, I headed toward the exit, feeling refreshed and grateful for the time spent outdoors. The park always has a way of making me feel at peace with the world.

1. Choose the correct answer from the alternatives. **1×5=5**

i. What is the main theme of the passage?

a) A visit to the park and the activities observed there.
b) A description of the weather conditions.
c) A detailed guide on how to enjoy nature.
d) A review of park facilities.

ii. How did the speaker feel after spending time in the park?

a) Tired and restless. b) Peaceful and relaxed.
c) Bored and uninterested. d) Annoyed by the noise.

iii. Which of the following sentences is correct in terms of subject-verb agreement?

a) The children was running around playing games.
b) The children were running around playing games.

c) The children run around playing games.
d) The children runs around playing games.
iv. In the sentence "I decided to stop and take a break under a large tree," which phrase is the object of the verb "decided"?
a) I   b) To stop   c) A break   d) Under a large tree
v. Which of the following sentences uses the correct punctuation?
a) The sun was shining brightly, and there was few clouds in the sky.
b) The sun was shining brightly and, there were few clouds in the sky.
c) The sun was shining brightly and there were few clouds in the sky.
d) The sun was shining brightly, and there, was few clouds in the sky.

2.	**Fill in the blanks using the most appropriate word from the options provided. Each blank can only be filled with one correct word.**			1×5=5

Last weekend, I decided to visit the (a)__________ to enjoy the beautiful weather. As I walked along the (b)_________, I saw several people relaxing, while others were engaged in (c)________ activities. Some were sitting on benches, reading or chatting, while a family was having a picnic near the (d)_________. A few people were riding (e)_________, enjoying the peaceful surroundings.

3.	**Identify nouns mentioned in the passage. (any three) Make 2 complex sentence by using those nouns·**	**CLO5**	**Apply**	3+2=5

Section B

4.	Explain the concept of the **Verb** in English. Provide examples of how to form finite and non-finite verbs.	**CLO1**	**Define**	5×1=5
5.	What is **Clause**? Explain simple, complex and compound sentence by providing examples.	**CLO1**	**Define**	5×1=5
6.	**Identify and correct the punctuation errors in the following passage: (10 errors)**	**CLO5**	**Understand**	0.5×10=5

the lead engineer said we must enhance the performance of the system.its current output is below expectations and we have already identified the weak points in the design. however, we still need to test the proposed solutions be careful the manager replied without proper testing we risk compromising the entire project timeline."

Hamdard University Bangladesh

Dept. of Computer Science & Engineering

Midterm Examination (Fall'24)

PHY 111 Physics

Time: 1.5 hours   Marks: 30

Answer any 3 (Three) of the following sets of questions

1	(a)	"Justify the statement: 'All Simple Harmonic Motion is Periodic, but All Periodic Motion is not Simple Harmonic Motion.' Provide appropriate examples to support your justification".	3	CLO1
	(b)	Show analytically that in SHM the acceleration is zero when the velocity is maximum and that the velocity is zero when the acceleration is maximum.	4	CLO2
	(c)	Find whether the discharge of a condenser through the following inductive circuit is oscillatory: $C = 0.1\mu F$, $L = 10mH$, and $R = 200\Omega$. If the circuit is oscillatory, calculate its frequency.	3	CLO3
2	(a)	State Hooke's Law. From this law derive the differential equation for Simple Harmonic Motion (SHM) and solve the equation to obtain an expression for the displacement of a particle executing SHM.	6	CLO1
	(b)	A student at HUB is investigating simple harmonic motion in a Physics Lab. An oscillating mass on a spring is used in the investigation. The student measures that a body of mass 0.5 kg is suspended from a spring, stretching it by 0.07 m. For a displacement of 0.03 m, the body has a downward velocity of 0.4 m/s. Compute the following: i) the time period of the oscillation. ii) the frequency of the oscillation. iii) the amplitude of the vibration of the spring.	4	CLO3
3	(a)	For a two body oscillator system, prove the relation, $T = 2\pi\sqrt{\frac{\mu}{K}}$ where, $\mu = \frac{m_1 m_2}{m_1+m_2}$ = reduced mass.	4	CLO2
	(b)	A SHM is represented by the equation, $x = 10\sin\left(10t - \frac{\pi}{6}\right)$ in SI units. Calculate: i) the frequency, ii) the time period, iii) the maximum displacement, iv) the maximum velocity, v) the maximum acceleration, and vi) displacement, velocity, and acceleration at time t = 1 second.	6	CLO3

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HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Batch: 27^{th} | Semester: Fall 2024
Time: 1 hour and 30 minutes
Date: 11/01/2025 | **Mid-Term Examination** | Total Marks: 30

Answer any **five** questions from the followings. 5x8= 40

1. (a) Define the term Subset and Cardinality of a set. CLO 4 2
   (b) Let U be a universal set that contains all numbers is less than 20 and V be a set that contains all Prime numbers is less than 13. Show the complement of V in Venn diagram. Find the power set of V and cardinality of the complement of set V. CLO 4 4

2. (a) Briefly explain the term predicate calculus and propositional calculus. CLO 1 2
   (b) Let $f_1(x) = x^3 - 2x^2 + 1$ and $f_2(x) = x^3 + 4x - 2$. Determine the function with reason. Find f_1f_2 and f_1+f_2. CLO 4 4

3. (a) Determine the general equation for the Tower of Hanoi Problem. CLO 4 3
   (b) What is meant by Recurrence relation? Find the first five terms of the sequence defined by the recurrence relation and initial condition given below: $\mathbf{a_n = na_{n-1} + n^2a_{n-2}, a_0 = 1, a_1 = 1}$. CLO 4 3

4. (a) Write short notes on Exclussive OR and Tautology of a Proposition. CLO 1 2
   (b) Prove De-Morgan's law for three bits by using proposition. CLO 1 4

5. (a) Write the properties of a relation. CLO 4 2
   (b) Let A = {1, 2, 3}, B = {1, 2, 3, 4} and C = {0, 1, 2} be 3 sets. R be a relation from A to B and S be a relation from B to C, where R = {(a, b) | $a \le b$} and S = {(b, c) | $b \le c$}. Find R, S, $S \circ R$, $R \circ S$. CLO 4 4

6. Prove that the expressions $[(p \land \neg((\neg p \lor (q \land \neg r)) \to q)) \lor (p \land q)] \to p$ and $\neg p \lor (p \land \neg q) \lor r \lor (q \land \neg r)$ are logically equivalent. CLO 1 6

7. Let A = {1, 2, 4, 12, 24, 48} be a set, find the following relations: CLO 4 6
   $R_1 = \{(a, b) \mid a <= b\}$,
   $R_2 = \{(a, b) \mid a > b\}$,
   $R_3 = \{(a, b) \mid a \text{ divides } b\}$ ad
   $R_4 = \{(a, b) \mid a^2 + b <= 200\}$
   Find the properties with reason of each relation. Represent each relation in matrix form and graphically.

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Midterm Examination (Fall'24)

EEE 117 Electrical Engineering

Time: 1.5 hours Marks: 30

Answer any 3 (Three) of the following sets of questions

1 (a) Define the following terms: 2 CLO1

I. Charge
II. Current
III. Voltage
IV. Power

(b) Calculate the power supplied or absorbed by each element in figure-1(b). 4 CLO3

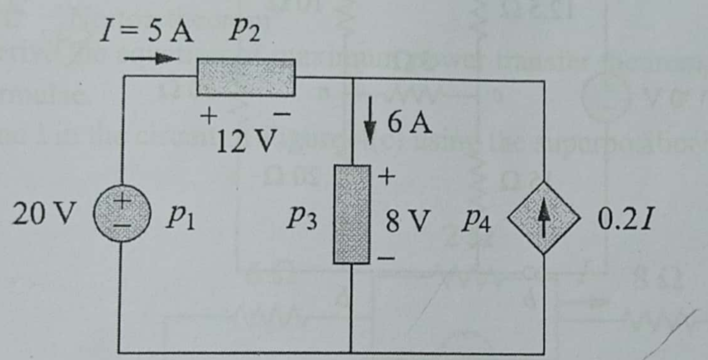

Figure-1(b)

(c) Calculate the equivalent resistance in the circuit in Figure-1(c). 4 CLO4

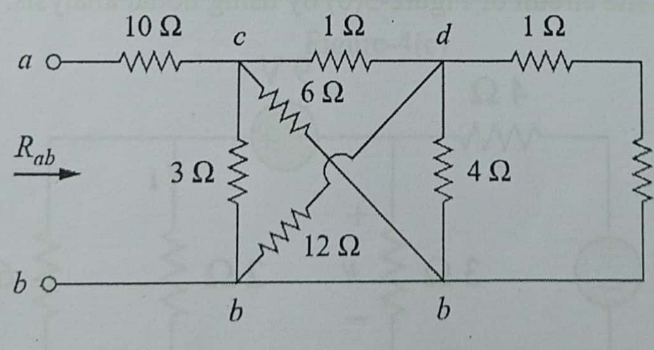

Figure-1(c)

2 (a) State Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL). 2 CLO1

(b) Find currents and voltages in the circuit shown in Figure-2(b). 4 CLO3

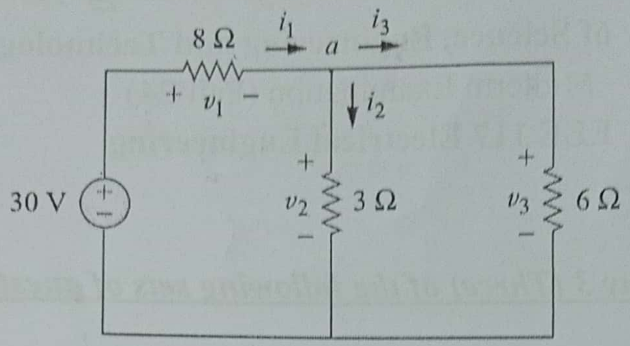

Figure-2(b)

(c) Obtain the equivalent resistance R_{ab} for the circuit in Figure-2(c) and use it to find current i. 4 CLO4

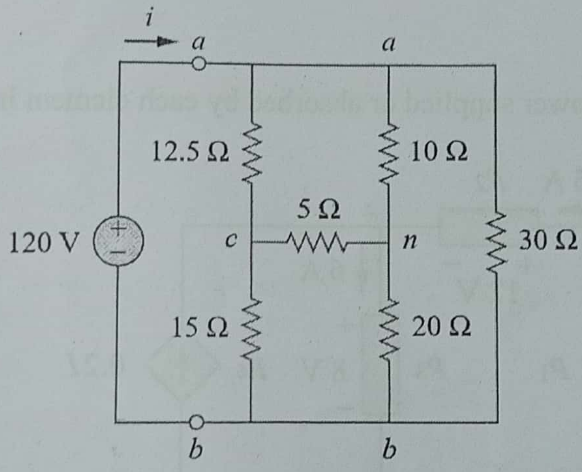

Figure-2(c)

3 (a) Write down the techniques of circuit analysis briefly. 2 CLO1

(b) Find v and i in the circuit of Figure-3(b) by using nodal analysis. 4 CLO3

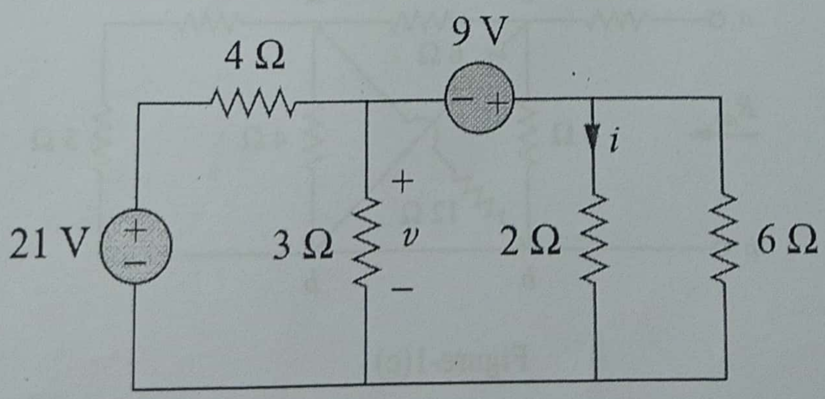

Figure-3(b)

(c) Use mesh analysis to find the current I_0 in the circuit of Figure-3(c). 4 CLO4

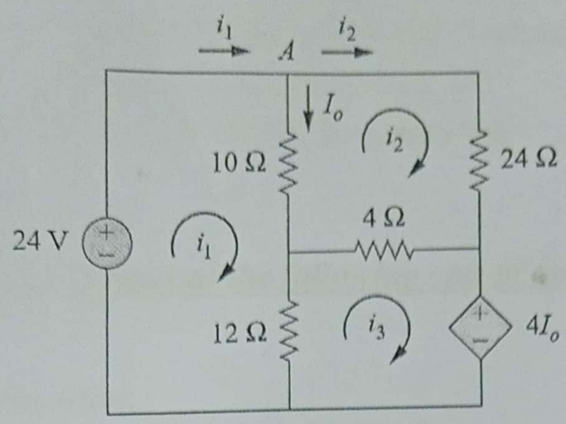

Figure-3(c)

4 (a) Write a short note on following terms. 2 CLO1

I. Thevenin theorem

II. Norton theorem

(b) Derive the equation of maximum power transfer theorem and note down the formulae. 4 CLO2

(c) Find I in the circuit of Figure-4(c) using the superposition theorem. 4 CLO3

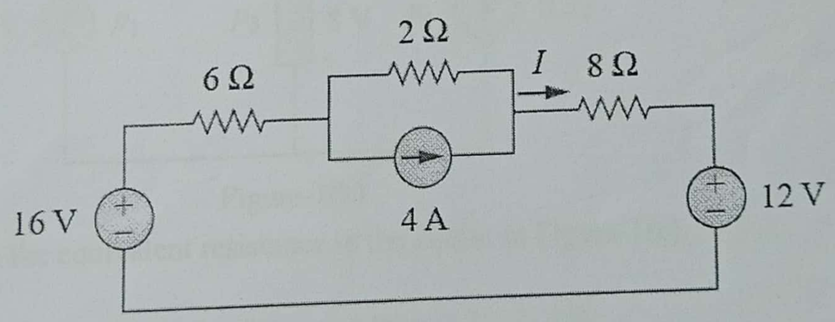

Figure-4(c)